Chapter I – Introduction

Machine Learning

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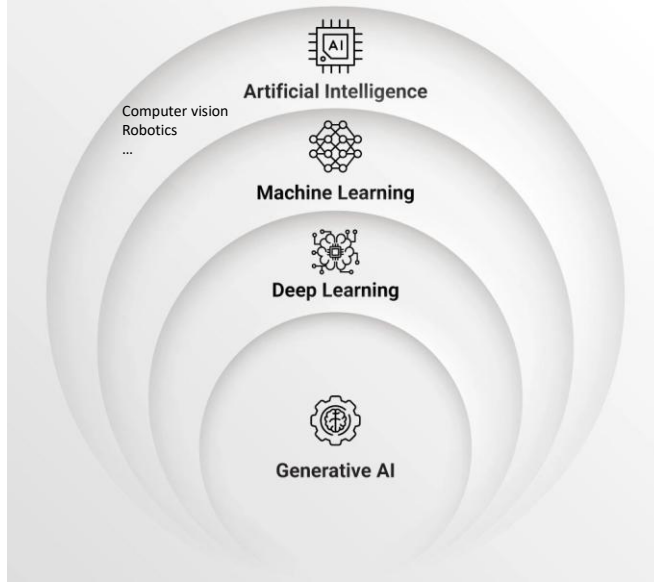
1

Outline

- Data Mining
- Statistical Learning
- Prediction / Inference
- Machine Learning
  - Types of Learning
    - Supervised / Unsupervised
  - Classification/Regression
  - Main Challenges
  - …

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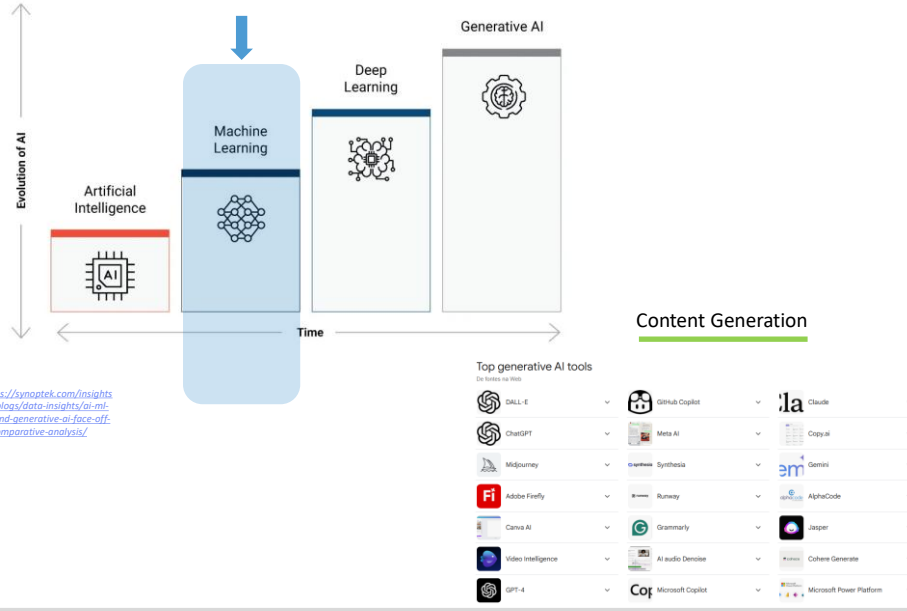
Introduction



https://synoptek.com/insights/it-blogs/data-insights/ai-ml-dl-and-generative-ai-face-off-a-comparative-analysis/

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Introduction



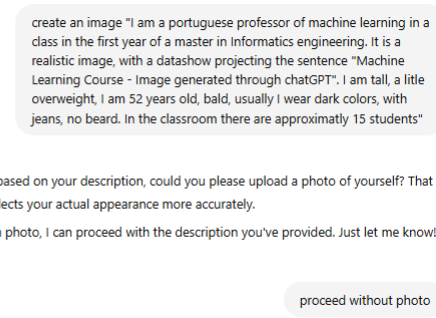
https://synoptek.com/insights/it-blogs/data-insights/ai-ml-dl-and-generative-ai-face-off-a-comparative-analysis/

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Introduction

- Generative AI







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Introduction

- Machine Learning vs Generative AI
  - Different Application areas
  - Different goals

Area	Generative AI	Machine Learning
Healthcare	drug discovery (molecular simulation, clinical trial optimization, etc)	disease diagnosis
Finance	personalized financial advice	fraud detection
Retail	product design	inventory management/alerts
Manufacturing	prototype design	predictive maintenance

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Generative AI

Some examples

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Introduction



I have two free days in London, suggest a plan to visit the top attractions

Day 1: Central London – History & Iconic Sights

Morning:

- **Buckingham Palace** (see the Changing of the Guard around 10:45am)
- Walk through **St. James's Park** to:
- **Westminster Abbey** (optional tour inside)
- See **Big Ben** and the **Houses of Parliament**

Lunch:

- Try a pub like *The Red Lion* (near Parliament) or *Walkers of Whitehall*

Afternoon:

- Walk along the **Thames River**, cross **Westminster Bridge**
- Visit the **London Eye** (panoramic city views)
- Stroll the **South Bank** – great for street performers, bookstalls, and cafes
- Option: Stop by **Tate Modern** (free entry)

Dinner:

- **Covent Garden** area – good mix of restaurants & lively atmosphere

Evening:

- Option: West End show (book in advance for a musical or play)
- Or take a **Thames night cruise** for views of the city lights

Day 2: Culture, Markets & Views

Morning:

- Visit **Tower of London** (arrive early, see Crown Jewels)
- Walk to **Tower Bridge** and explore the bridge exhibition
- Stroll to **Borough Market** for coffee/snacks

Lunch:

- Eat at **Borough Market** – wide selection of global street food

Afternoon:

- Visit **The British Museum** or **National Gallery** (both free)
- Walk through **Soho** or **Carnaby Street** for shops and people-watching

Optional add-on:

- Take a short tube ride to **Sky Garden** (free, book ahead) or **The Shard** (paid) for skyline views

Dinner:

- Head to **Shoreditch** or **Soho** for trendy restaurants
- If you want traditional: *Rules* (Covent Garden) or *Dishoom* for modern Indian

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Introduction



describe the image

The image shows a **bright yellow compact car** set against a vibrant, gradient background with hues of orange, purple, pink, and blue, giving it a futuristic and energetic look. The car is a **Renault 5 EV prototype**, a modern electric version inspired by the classic Renault 5 from the 1970s and 80s.

Key details visible:

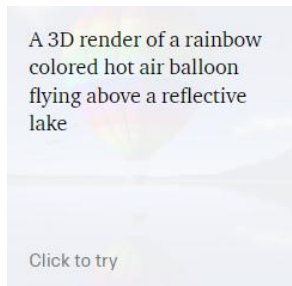
- Bold and sleek styling with modern LED headlights
- Renault badge on the front grille
- Sporty wheels and a compact urban hatchback profile
- Front license plate reads "RENAULT 5"

This image seems to be part of a promotional or concept reveal, emphasizing the car's design and electric innovation in a stylish, eye-catching setting.

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Introduction

- Image generator from a description in natural language
  - https://openai.com/index/dall-e-2/





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Introduction

- Midjourney (https://www.midjourney.com/explore?tab=top)

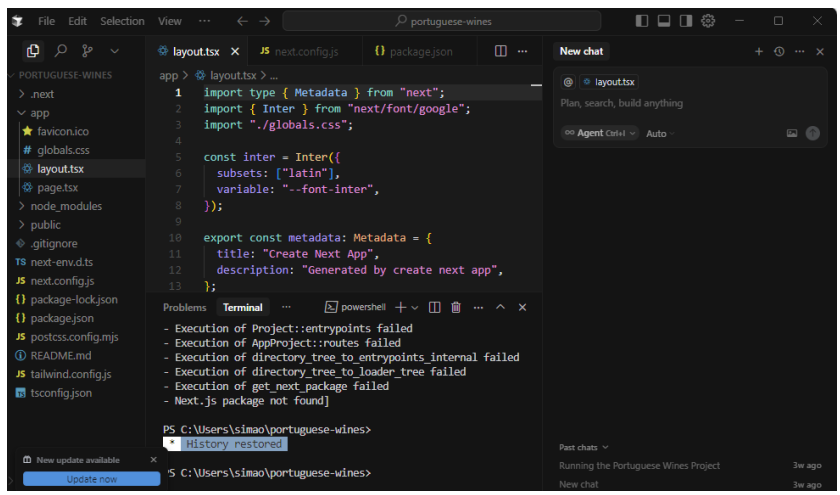


Hot air balloon created by huge white crocs rubber slippers, with "crocs" written on it, girl eating watermelon, white clouds in the sky, looking down is the sea, people swimming party, surreal interesting summer atmosphere advertisement, real photography, very realistic, 64k high definition, shocking creative vision

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Introduction

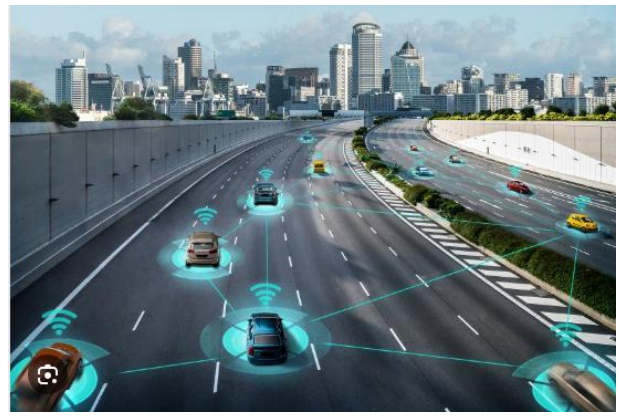
- Cursor (https://www.cursor.com/); Copilot (https://copilot.microsoft.com); Vercel (https://vercel.com/)



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Introduction

- Autonomous Driving
  - Simulation and Synthetic Data Generation



https://www.ourcrowd.com/investment-themes/autonomous-cars

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Machine Learning

Some examples

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Introduction

- Disease Detection





https://binariks.com/blog/ai-machine-learning-for-early-disease-detection/

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Introduction

- Spam filter



https://mailsafi.com/blog/best-email-filter-cannot-guarantee-100-protection/

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Data Mining

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Data Mining

- From Data to Decision
  - Data drives the world !!
    - This the era of information
    - Every second new data is being generated and stored



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Data Mining



Data

Information

Knowledge

Adapted from "Gestão de dados de investigação e o plano de gestão de dados", Bruno Direito, CISUC, junho, 2024

Data Mining

- Data
  - Unit fact / unit element / measurement

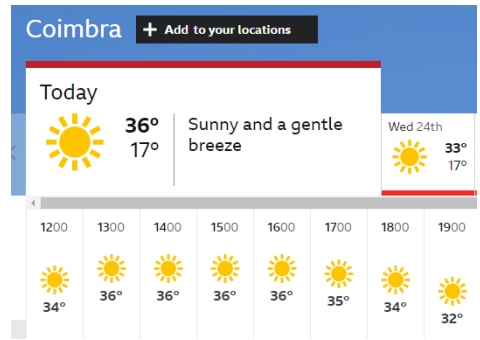


- Dataset
  - Set of data that share a context



Data Mining

- Information
  - Data + Context (metadata)
    - where
    - when
    - units
    - who created the data
    - …



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Data Mining

- Knowledge
  - Information + understanding → support to decision-making
    - e.g. type of clothes, sun protector, …

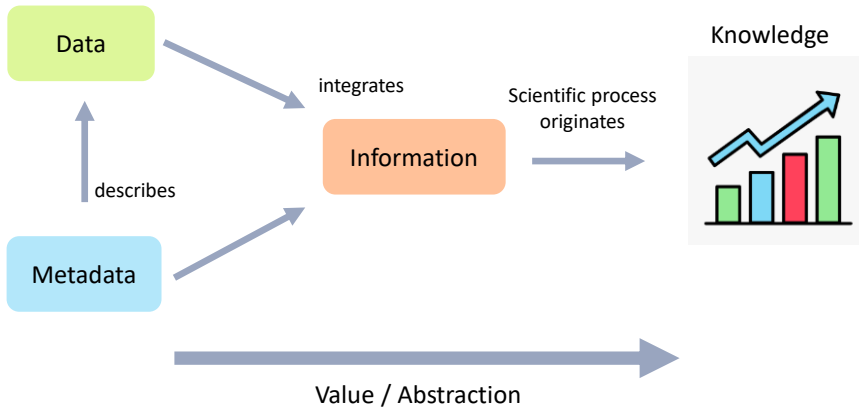


https://kg2.com.au/satellite-based-weather-forecasting-for-agricultural-decision-making/

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Data Mining

Data / Information / Knowledge



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Data Mining

- New challenges / Opportunities
  - Large Data Sets are collected /available
    - Meaningful/useful information has to be extracted from data!
      - Otherwise, those datasets are **useless!**

Challenge!



Opportunities!

  - Development of data analysis algorithms / modelling techniques to:
    - derive real insights from the data
    - support professional decision making (e.g. clinical decision, …)
    - achieve better decision, improve the strategic vision and revenues of companies
    - …

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Data Mining

- e.g.: Health Care

Context:

Social, Clinical, Economic



- Population is aging
- Increases in life expectancy and decreases in birth rates
- Prevalence of chronic diseases (e.g. cardiovascular and neurological diseases)
- Great burden on health systems

National Health Systems tend to be unsustainable ⇨ Need to **reduce costs**

Opportunity for new paradigms for health care delivery systems

Prevention, Health Care Optimization

[support to clinical decision]



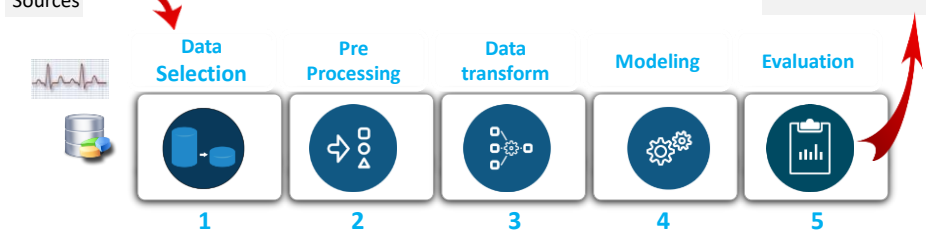
Data Mining

- From Data to Decision
  - e.g. Support to Clinical Decision

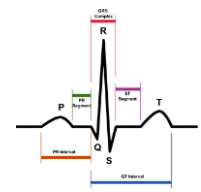
Data Sources



Decision
- Stratification
- Diagnosis
- Prognosis

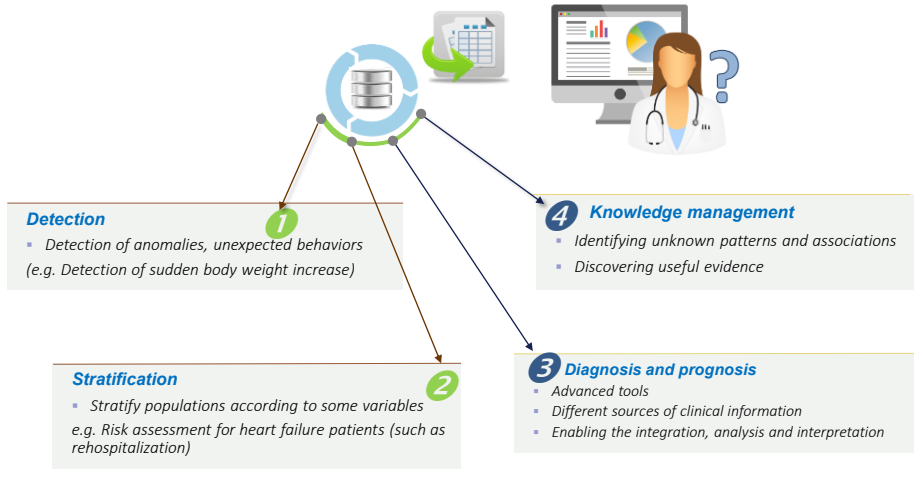


1. Select data from various sources
2. Prepare and clean the data
3. Map complex objects (e.g. time series) to features
4. Patterns, classification, rules
5. Results and visualization



Data Mining

- Support to Decision (e.g. Clinical Context)
  - Extraction of meaningful information from clinical Datasets



Data Mining: CRISP-DM

- **CR**oss **I**ndustry **S**tandard **P**rocess for Data Mining (CRISP-DM) is a process model that describes the data science life cycle:
  - Composed of 6 main phases

1. Business understanding: clear understanding of business challenges

2. Data understanding: What data is available? Is it clean?

3. Data preparation: prepare data to apply models

4. Solution: models/solutions to meet the requirements

5. Evaluation: test the results against the goals of the project

6. Deployment —deploy the solution and/or results

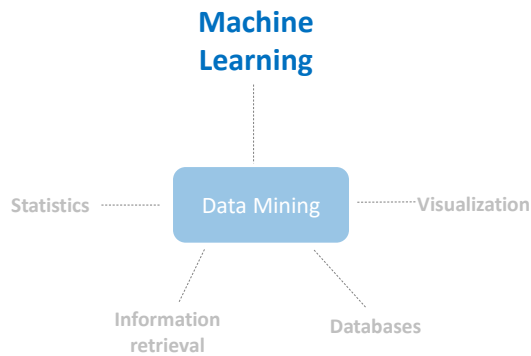


https://www.researchgate.net/figure/CRISP-DM-data-mining-framework_fig1_341627969

Data Mining

Data Mining is the process of extraction useful information (patterns and knowledge) from (often large) datasets.

- Convergence of multiple disciplines



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Data Mining

- New applications
  - Finances
  - Health
  - Industry
  - Digital marketing
  - …



https://multicursos.com.br/blog/o-que-e-big-data-e-como-tirar-proveito-dessa-tecnologia

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Statistical Learning

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Statistical Learning

- Develop a **model** that can be used to **predict sales** based on:
  - 3 media budgets (TV, radio, newspapers)



https://www.crmsoftwareblog.com/2020/05/april-2020-new-feature-deep-dive-sales-forecasting-webinar/

$$Y = \hat{f}(X) + \epsilon$$

Y - *sales*

$X = (X_1, \ldots, X_p)$ *media budgets*

p *different predictors*

ϵ - *random error term*

- **Goal:** Maximize the revenue!!

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Statistical Learning

- Input Variables / **Features**

X_1	X_2	X_3
TV budget	Radio budget	Newspaper budget

- Output Variables / Response

Y
Sales

$$Y = \hat{f}(X) + \epsilon$$

Statistical Learning

- Considering:

$$Y = f(X) + \epsilon$$

  - f *is a* function of $X_1, \ldots, X_p$
    - represents the systematic information that X provides about Y
  - ϵ is a random error term
    - independent of X probably due to unmeasured variables, ….

> *" Statistical Learning refers to a set of approaches for estimating f."*

  - $\hat{f}$ why?
    - The estimation of f allows **prediction** and **inference**

Prediction / Inference

X_1	X_2	X_3
TV budget	Radio budget	Newspaper budget

Y
Sales

$$Y = \hat{f}(X) + \epsilon$$

- Prediction
  - Model predicts the outcome for new instances.
- Inference
  - Apply the model to learn about the relationship between independent variables and the outcome (interpretability)

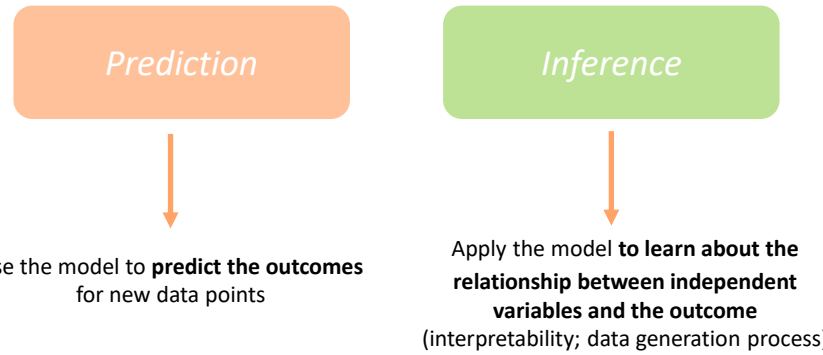
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Prediction / Inference

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Prediction / Inference

- Both approaches involve a learning process from data (supervised)
- Goal:
  - to derive a model that describes the relationship between the independent variables and the outcome
    - However, in two very different perspectives:



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Inference

Inference

- Determine how Y is affected by changes in $X_1, \ldots, X_p$
  - The estimation of f aims to clarify the relationship between X and Y, i.e. to understand how Y changes as a function of $X_1, \ldots, X_p$

$$\hat{Y} = \hat{f}(X)$$

  - In this case $\hat{f}$ **cannot be seen as a black box**, because the relationship between variables and the respective response is the main focus.

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Inference

- Frequent questions that can be answered:
  - Which predictors are associated with the response?
    - The identification of a limited number of predictors (most important) can **be extremely useful** (reduction of dimensionality)
  - What is the relationship between the response and each predictor?
    - Increasing the predictor increases/decreases the response?
  - Can the relationship between Y and each predictor be modeled using a linear equation?
    - More complicated expression?

Prediction

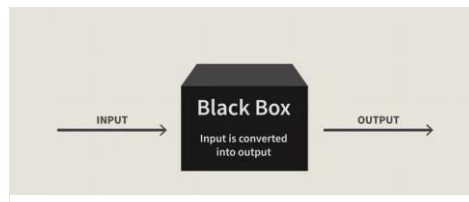
Prediction

Predict the outcomes for new data points.

$$\hat{Y}=\hat{f}(X)$$

where $\hat{f}$ is the f estimate and $\hat{Y}$ represents the Y prediction.

- In some conditions $\hat{f}$ can be considered as a **black box**, i.e. the exact form of $\hat{f}$ is not so important, provided that it assures accurate predictions for Y



https://www.investopedia.com/terms/b/blackbox.asp

Prediction



$X_1, \ldots, X_p$
Characteristics of a patient's blood sample easily measured in a lab

Y

Encoding the patient's risk for a severe adverse reaction to a particular drug

Predict Y using X is very useful as patients at high risk of severe adverse reaction must not take that specific medication

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Prediction

- The accuracy of $\hat{Y}$ as a prediction of Y depends on two values:
  - Reducible error
  - Irreducible error

$$E(Y - \hat{Y})^2 = \underbrace{[f(X) - \hat{f}(X)]^2}_{\text{reducible error}} + \underbrace{Var(\epsilon)}_{\text{irreducible error}}$$

where:

- $E(Y - \hat{Y})^2$ represents the average difference of the squared diferences between the predicted and actual value of Y,
- $Var(\epsilon)$ the variance associated with term ϵ

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Prediction

$$E(Y - \hat{Y})^2 = \underbrace{[f(X) - \hat{f}(X)]^2}_{\text{reducible error}} + \underbrace{Var(\epsilon)}_{\text{irreducible error}}$$

Reducible error

It is reducible so $\hat{f}$ can be improved

Irreducible error

Even if it were possible to achieve a perfect estimation of $\hat{f}$, the prediction of $\hat{Y}$ would still have some error due to ϵ

It is irreducible because no matter how well the estimate of f is performed, the error introduced by ϵ (unmeasured variables, unmeasurable variation, …) cannot be reduced.

Prediction/Inference ?

- Example: marketing campaign
  - **Goal:** to identify individuals who will respond positively to an email based on demographic variables measured on each individual
  - **predictors / features:** demographic variables (e.g. age, race, ethnicity, gender, income, education, …)
  - **outcome:** response to campaign (positive/negative)
  - The marketing company wants an accurate model to predict the response based on predictors.
    - The company **doesn't seek for an in depth understanding of the relationship between several predictors and outcomes**



https://www.searchenginejournal.com/launching-influencer-marketing-campaigns-key-considerations/316639/

Prediction!

Prediction/Inference ?

- Example: marketing campaign
  - **Goal:** modeling a brand of a product
  - **predictors:** price, store location, discount level, …
  - **outcome:** influence of each predictor in the probability of purchase
    - e.g. Which is the effect of changing the price on sales?

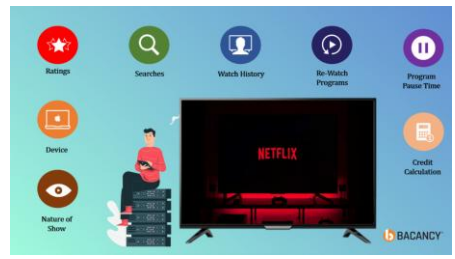


Inference!

https://www.searchenginejournal.com/launching-influencer-marketing-campaigns-key-considerations/316639/

Prediction/Inference ?

- Netflix Prize
  - A competition for the best recommendation system



https://www.linkedin.com/pulse/netflixs-recommendation-engine-artificial-brain-built-mahendra-kadam-uukif

Netflix utilizes machine learning algorithms to analyze user data and movie ratings, creating 1,300 recommendation clusters based on viewing preferences. This personalized approach ensures that each user is presented with a tailored list of movies and TV shows upon accessing Netflix. The objective is to help users discover content they will enjoy within a short timeframe, as users typically have just 90 seconds for this decision. While the recommendation system operates in the background, its impact is evident in the customized viewing experience for users.

Prediction/Inference ?



Build a recommendation system (machine learning algorithm) dependent on the star rating system **to identify what movies users might like to watch.**

Prediction

Build an interpretable **model to better understand why** people choose a specific movie.

Inference

Prediction/Inference ?

- Example: Real Estate
  - Variables:
    - crime
    - shops
    - public transportation
    - schools
    - air quality
    - views, ….



How much more will a house be worth **if it has a view of the river**?

Inference!

Given its characteristics determine if a **house is under or over valued***?*

Prediction!

Machine Learning

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Machine Learning

*"A computer program that automatically improves through experience, i.e. learns from experience, **E**, with respect to some task, **T**, and some performance measure, **P**, if its performance on **T** as measured by **P** improves with experience **E**. "* **[2][3]**



Spam filter:

- Given examples of spam mails (flagged by users) and examples of regular mail (no spam) **the filter can learn how to classify email**.
- Each training example is called training instance or sample
- the set of instances is designated by training set.

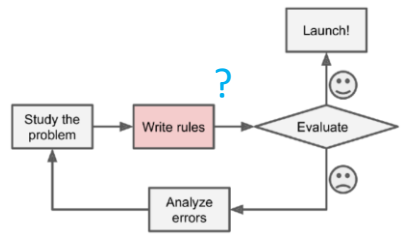
Spam filter:
T – flag spam for new mail
E – training data set
P – e.g. accuracy

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Machine Learning

- Traditional approach
  - identify patterns/words
  - **write rules** according to the previous identification
  - evaluate until the algorithm is ready for launching



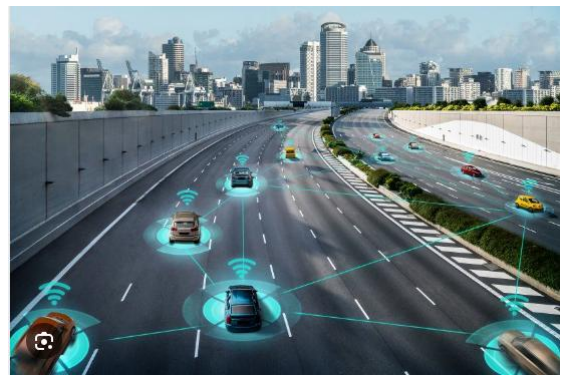


Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems

  - complex rules, hard to maintain, …

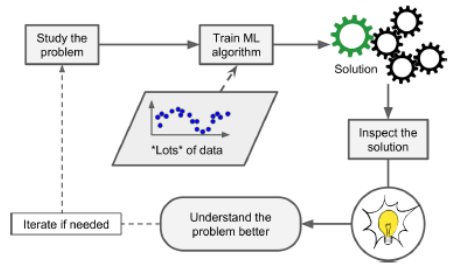
Machine Learning

- Traditional approach
  - Impossible to apply to complex/dynamic problems …



Machine Learning

- ML approach



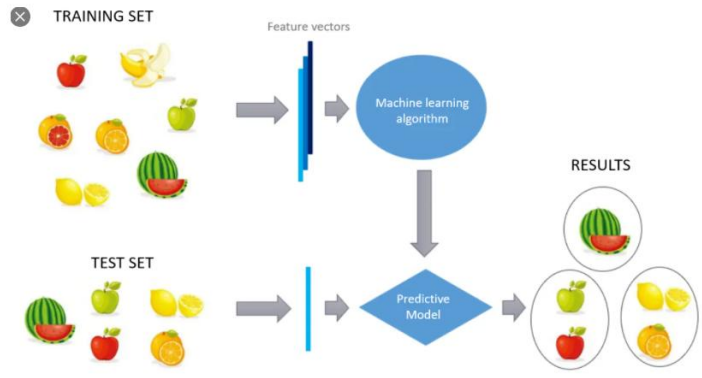
Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems

- **Can learn automatically from data and adapt to the specific conditions**, particularly well adapted to:
  - Problems with long list of rules (often are simplified with better performance);
  - Fluctuating environment as the system can adapt to new data;
  - Get insights about complex problems and large amount of data

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Machine Learning

*" … **building models** which, to be useful, **rely on a collection of examples of some phenomenon …**"*



https://www.diegocalvo.es/en/learning-non-supervised/

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Machine Learning

- Detecting card credit fraud



https://sdk.finance/credit-card-fraud-detection-big-players-experience/

- Image Classification



https://www.freecodecamp.org/news/chihuahua-or-muffin-my-search-for-the-best-computer-vision-api-cbda4d6b425d/

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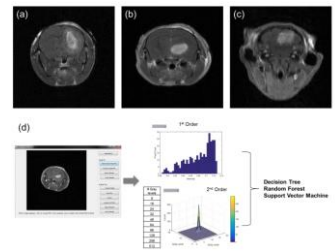
Machine Learning

- Object Detection





- Tumor detection



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Statistical Learning vs. Machine Learning

"The major difference between machine learning and statistical learning is their purpose. ***Machine learning models are designed to make the most accurate predictions*** *possible. Statistical models are designed for inference about the relationships between variables."*

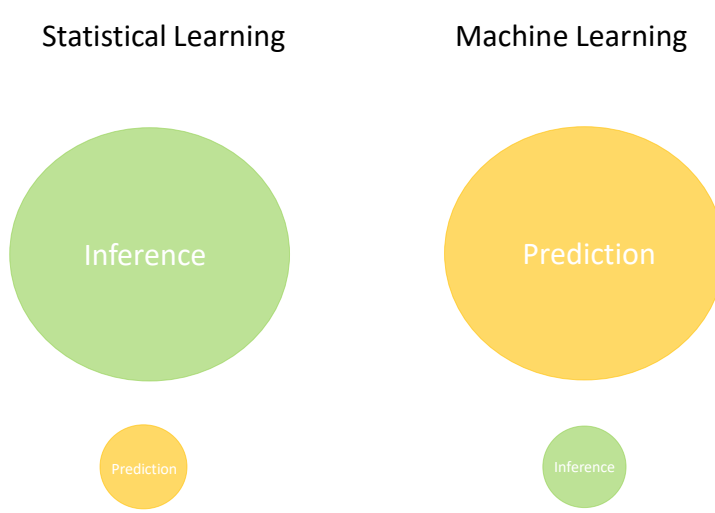
https://towardsdatascience.com/the-actual-difference-between-statistics-and-machine-learning-64b49f07ea3

MACHINE LEARNING	STATISTICAL LEARNING
Subfield of Artificial Intelligence	Subfield of mathematics
Uses algorithms	Uses equations
Requires minimum human effort; is automated	Requires a lot of human effort
Can learn from large data sets	Deals with smaller data sets
Has strong predictive abilities	Gives a best estimate: you gain some insights into one thing, but it's of little or no help with predictions
Makes predictions	Makes inferences
Learns from data and discovers patterns	Learns from samples, populations, and hypotheses

https://steelkiwi.com/blog/what-is-machine-learning/

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Statistical Learning / Machine Learning



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Machine Learning

Types of Learning

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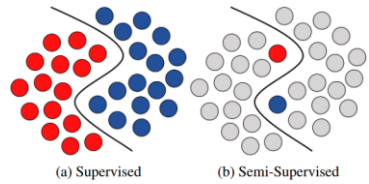
Machine Learning: Types of Learning

- Machine Learning algorithms can be classified according to the amount and type of supervision they get during the training phase:
  - Supervised Learning
    - The training data set includes the desired solutions (labels)
  - Unsupervised Learning
    - The training data is unlabeled
  - Semi-supervised Learning
  - Reinforcement Learning

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Machine Learning: Types of Learning

- Semi-supervised Learning
  - Few labeled instances combined with a lot of unlabeled instances.

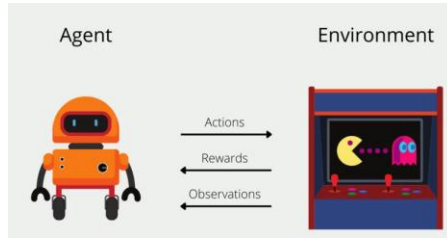


https://dasha.ai/en-us/blog/self-supervised-machine-learning

  - Semi-supervised learning may be applied when there are large amounts of unlabeled data/datasets.
    - A possible approach may rely on the manual labeling of data to use it as the training set for the model.
    - Once trained, the model can make predictions on the remaining unlabeled instances.

Machine Learning: Types of Learning

- Reinforcement Learning
  - The learning system observe the environment, select and perform actions and get rewards (positive/negative) in return
    - T**here is no external training data**! Actually, the algorithm works **exclusively based on a rewards system**.
    - It involves an autonomous agent that observes the environment and then selects an action that will lead to rewards, which help the algorithm to improve on its own.



https://databasecamp.de/en/ml/reinforcement-learnings

Machine Learning: Types of Learning



https://www.aitude.com/supervised-vs-unsupervised-vs-reinforcement/

Criteria	Supervised ML	Unsupervised ML	Reinforcement ML
Definition	Learns by using labelled data	Trained using unlabelled data without any guidance.	Works on interacting with the environment
Type of data	Labelled data	Unlabelled data	No – predefined data
Type of problems	Regression and classification	Association and Clustering	Exploitation or Exploration
Supervision	Extra supervision	No supervision	No supervision
Aim	Calculate outcomes	Discover underlying patterns	Learn a series of action

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Machine Learning

Types of Learning

Supervised Learning

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Machine Learning: Supervised Learning

- The dataset is a collection of **labeled** examples:

$$\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{N}$$

  - Each element $\mathbf{x}^{(i)}$ among N is called a feature vector/**instance**
    - Each dimension of an instance j $= 1, \ldots, \mathrm{D}$ contains a value $x_j^{(i)}$ (feature).
    - For all N instances, the feature at position j always contains the same kind of information.

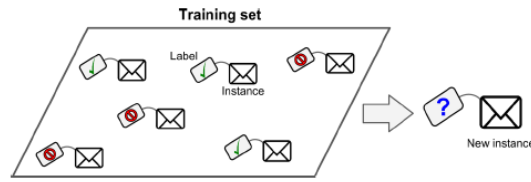


id	x_1 Gender	... ?	x_D Weight	y Label
1	$x_1^{(1)}$	...	$x_D^{(1)}$	$y^{(1)}$
...				
N	$x_1^{(N)}$	...	$x_D^{(N)}$	$y^{(N)}$

  - $y^{(i)}$ can be an element of a finite set of classes $\{C_1, \ldots, C_M\}$, a real number or a more complex structure (vector; multi-label classification)

Machine Learning: Supervised Learning

- The goal of supervised learning is to use a **complete labeled dataset** to produce a **model** that based on an input $\mathbf{x}$ (feature vector) can output the **label** for that **input**.

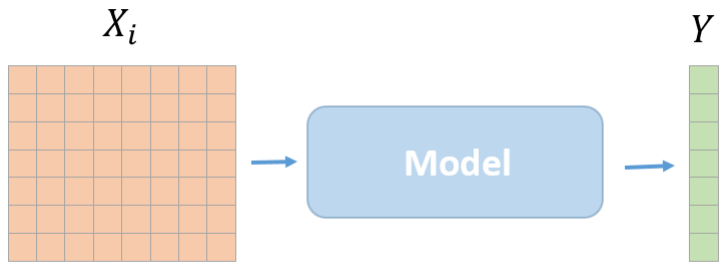


  - for each instance $\mathbf{x}^{(i)}, i = 1, \ldots, \mathrm{N}$ there is an associated response measurement $y^{(i)}$
  - it allows to build a model that **can predict the label** of a future (new) instance

Supervised Learning : Example

- Spam Filter

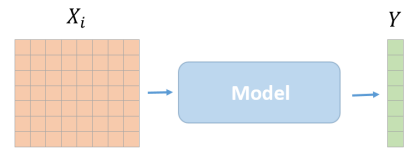




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Supervised Learning : Example

- Spam Filter
  - 1. Gathering the data
    - Collection of pairs (input, output)
      - 10000 labeled messages
      - 2 classes {“spam”; “not_spam”}



  - 2. Prepare the data
    - Feature vector
      - Data analyst decides to create a set of words (in this specific case called a “bag of words”) formed by 20000 words.
      - For each message, a feature vector is created (1 means the word exists; 0 means the word does not exist)
        - Thus, 10000 binary feature vectors (each one with dimension $\mathrm{D} = 20000$) and a label (“spam”/”not_spam”)

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Supervised Learning : Example

- 2. Prepare the data
  - Feature vector



Instances (messages)		Features (words) x_1	...	x_{20000}	Label (0 - not_spam / 1 - spam) y
	$\mathbf{x}^{(1)}$	0		1	0
	...		...	...	...
	$\mathbf{x}^{(10000)}$	1		1	1

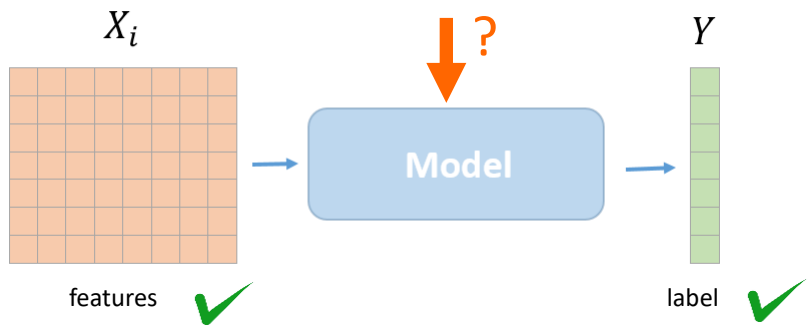
- Next step:
  - the label has to be transformed to machine-readable data. In this case (as the classifier is the support vector machine (SVM)):
    - "spam" situation is mapped to 1
    - "not_spam" to -1

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Supervised Learning : Example

- Spam Filter





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Supervised Learning : Example

- 3. Apply Training Data to a Learning Algorithm
  - For instance the algorithm is Suport Vector Machine (SVM)
  - SVM sees each feature vector as a 20000 dimension vector and **builds a hyperplane that separates positive and negative labels**
  - The SVM hyperplane is given by:

$$\mathbf{wx} + b = 0$$

where:

$\mathbf{w}$ is a real value based vector of the same dimensionality as $\mathbf{x}$

b is a real number

$\mathbf{wx}$ is the dot product and it is given by:

$$\mathbf{wx} = w_1 x_1 + w_2 x_2 + \ldots + w_D x_D$$

$D = 20000$ is the dimension of vector $\mathbf{x}$.

Supervised Learning : Example

- The predicted label for a given input vector $\mathbf{x}$ is given by:

$$y = sign(\mathbf{w}^* \mathbf{x} + b^*)$$

where $sign$ returns:

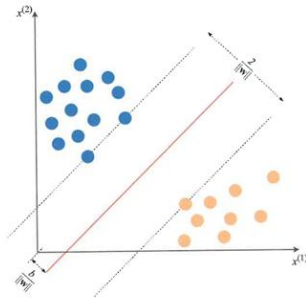
  - +1 if the argument value is positive
  - -1 if the argument value is negative.

- Based on the dataset, the **goal** of **the learning algorithm**, is:
  - to find the **optimal values** of $\mathbf{w}^*$ and b^* that can provide a reliable prediction (+1/"Spam"; -1/"not_spam") for a given feature vector $\mathbf{x}$

Supervised Learning : Example

- How to find $\mathbf{w}^*$ and b^*?
  - Optimization problem solved by the SVM algorithm under constraints:

$$\mathbf{wx}^{(i)} + b \geq +1 \quad if \quad y^{(i)} = +1$$
$$\mathbf{wx}^{(i)} + b \leq -1 \quad if \quad y^{(i)} = -1$$

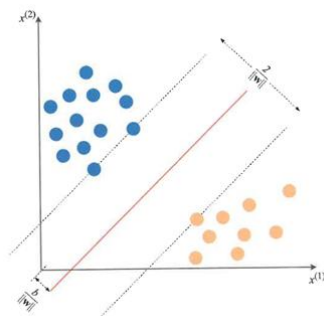


- Blue dots represent positive examples and orange the negative ones.
- $\mathbf{wx} + \mathrm{b} = 0$ is the decision boundary.

Supervised Learning : Example

- The hyperplane should separate the 2 classes with the largest margin

The higher the margin the better generalization!



Supervised Learning : Example

- In order to achieve that aim (high margin) the Euclidean norm of $\mathbf{w}$ must be minimized

$$\|\mathbf{w}\| = \sqrt{\sum_{j=1}^{D} (w_j)^2}$$

- So, the global **optimization problem** can be stated as:

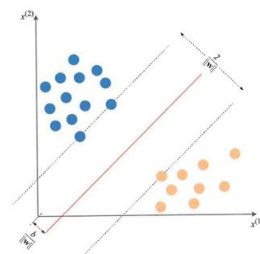
Minimize $\|\mathbf{w}\|$ subject to $y^{(i)} sign\left(\mathbf{w}\mathbf{x}^{(i)} + b\right) > 0 \quad for \;\; i = 1, \ldots, N$

  - Only accept w and b values such that every training example is correctly classified.

- The solution of this optimization problem (determination of $\mathbf{w}^*, b^*$) is called **model**, while **training** is the process of buiding the model.

Supervised Learning : Example

- Why is the previous model able to predict correctly unseen examples?
  - IF the examples applied for training:
    - were selected randomly
    - are independent of each other
    - follow the same procedure

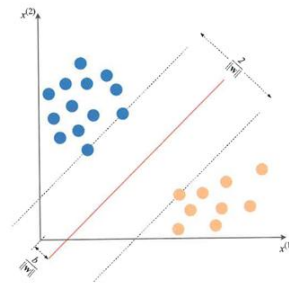


THEN

  - Statistically is more likely that:
    - a negative example will be located somewhere not too far from other negative examples.
    - positive example will be located somewhere not too far from other positive examples.

Supervised Learning : Example

- Decision boundary probably will separate new positive and negative examples from one another.
  - The **larger the training set**, the more unlikely new examples will be dissimilar from the examples used for training.
  - Large margin means less prone to classification errors



Further reading: PAC (probability approximately correct) is a framework for mathematical analyses that aims to identify under what conditions a learning algorithm will probably output an Approximately Correct Classifier.
https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118023471.ch11

Supervised Learning

- To keep in mind:
  - **Any classification learning algorithm** implicitly or explicitly **creates a decision boundary.**
    - In this context, the solution of an optimization problem is called **model**, while **training** is the process of buiding the model.
    - Other parameters that should be taken into account to classifiers selection:
      - Speed of model building (training)
      - Prediction processing time
      - …

Machine Learning

Types of Learning

Supervised Learning
Unsupervised Learning

Unsupervised Learning

- The dataset is a collection of **unlabeled** examples:

$$\{(\mathbf{x}^{(i)})\}_{i=1}^{N}$$

  - The system tries to learn without label information

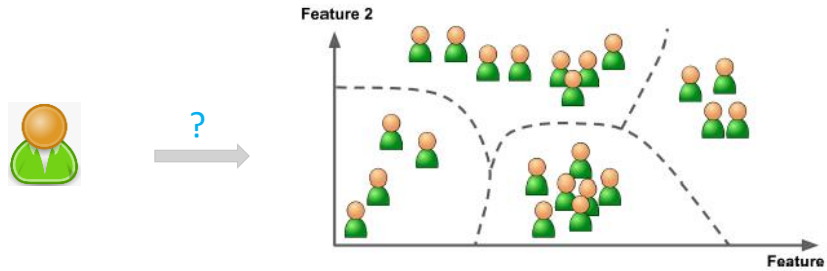


id	x_1 Gender	... ?	x_D Weight	y Label
1	$x_1^{(1)}$	...	$x_D^{(1)}$	$y^{(1)}$
...				
N	$x_1^{(N)}$	...	$x_D^{(N)}$	$y^{(N)}$

  - the goal is to create a model that takes a feature vector $\mathbf{x}^{(i)}$ as input and transforms it into another vector/value that can be used to solve a practical problem.

Unsupervised Learning

- e.g. Clusters
  - create groups of people according to the features of $\{(\mathbf{x}^{(i)})\}_{i=1}^{N}$ and of course with the specific problem
    - The goal of cluster analysis is to find out based on $\{(\mathbf{x}^{(i)})\}_{i=1}^{N}$ different groups so that the instances fall into relatively distinct groups.
      - With real data and complex problems it is very likely that some overlap can occur between groups.
    - Applied to customers preferences, health status, ….



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Unsupervised Learning

- It is applied across several real world domains:
  - Recommendation systems
  - Customer segmentation
  - …



https://www.linkedin.com/posts/abhiram-yadav-myla-7b9882147_machinelearning-unsupervisedlearning-segmentationandrecommendation-activity-7313369778725744640-Bfn2

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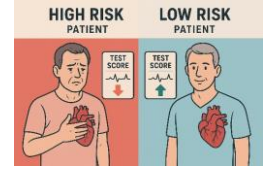
Machine Learning

Types of Learning
Classification/Regression

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Classification/Regression

- Variables can be qualitative or quantitative
  - A qualitative variable refers to values that belong to K different classes/categories:
    - spam/not_spam;
    - high_risk/low_risk;
    - male/female;
    - brand_A/ brand_B/ brand_C
    - …



  - A quantitative variable assumes numerical values, such as:
    - person's age
    - income
    - house price, ….

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Classification

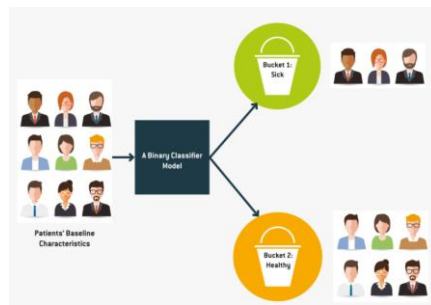
- In machine learning, the **classification** problem is solved by a classification learning algorithm that
  - takes a collection of labeled examples as inputs and **creates a model**
  - that model takes an unlabeled example and **automatically assigns a label to it**
    - e.g. Credit card fraud detection (binary classification – 1/0)



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Classification

- Classification
  - Finite set of classes
  - If there are only 2 classes (e.g. yes/no, male/female, …) it is a binary classification
  - If the number of classes is ≥ 3 it is a multiclass classification
    - There are some strategies to transform a binary classification into a multiclass classification



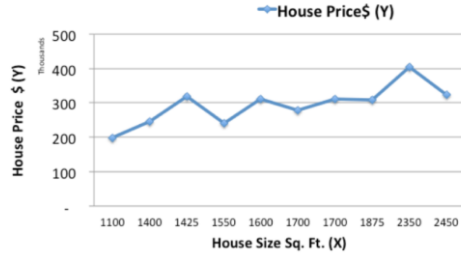
https://www.pngitem.com/middle/hRJJRJx_machine-learning-binary-classifier-hd-png-download/

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Regression

- Regression
  - predict a target numeric value (continuous value) given an unlabeled instance

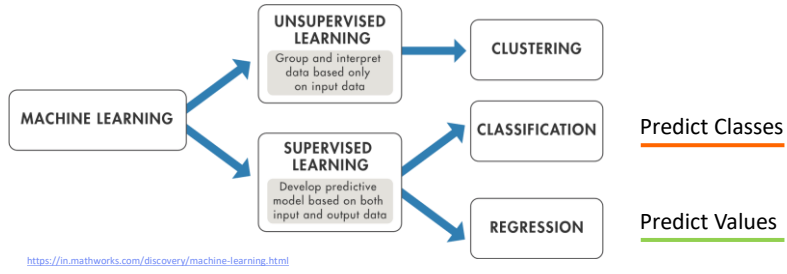




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Classification/Regression

- To keep in mind!
  - Supervised learning:
    - Trains a model on known input and output data (labeled data) so that it can predict future outputs
  - Unsupervised learning:
    - which finds hidden patterns or intrinsic structures in input data (unlabeled data).



https://in.mathworks.com/discovery/machine-learning.html

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Machine Learning

Types of Learning
Classification / Regression
Model based / Instance-Based Learning

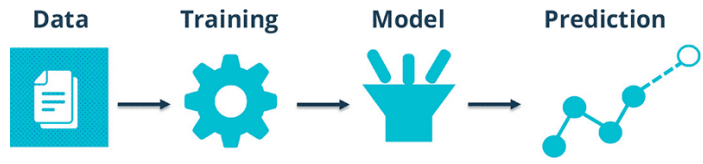
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Model based/Instance-Based Learning

- The majority of supervised algorithms are **model based**
  - Model based (e.g. SVM)

$$y = sign(\mathbf{w}^*\mathbf{x} + b^*)$$

    - Model parameters were learned from the training data.
    - After the model is built **the training data may/should be discarded.**

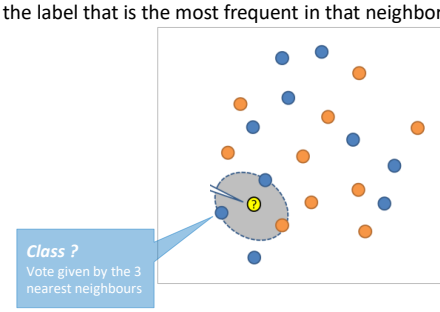


https://www.ram-ai.com/machine-learning/

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Model based/Instance-Based Learning

- Instance-based algorithms
  - A model is not created
  - KNN - Nearest Neighbor
    - To predict a label for an unseen example the KNN algorithm
      - all the instances are considered in each prediction
      - looks at the close neighborhood of the input in the space of feature vectors
      - outputs the label that is the most frequent in that neighborhood.



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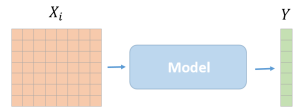
Machine Learning

Types of Learning
Classification / Regression
Model based / Instance-Based Learning
Batch / Online Learning

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Batch Learning

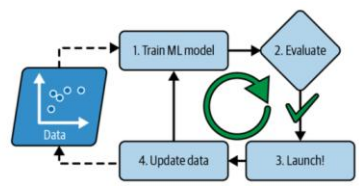
- The ability to learn incrementally is another criterion to characterize Machine Learning systems:
  - Batch Learning
    - The system **is not able to learn incrementally**, it must be trained using all available data.



    - System is trained and then runs without learning anymore (offline learning)
      - Update with new data requires a new training phase from scratch but taking old and new data
      - The whole process of training, evaluating and launching a new classifier can be automated
    - Simple solution
    - In some situations it can be time consuming and may require a lot of resources
      - CPU/GPU, RAM, disk space, …

Batch Learning

- Batch learning can be performed at regular intervals:
  - Weekly, monthly, …

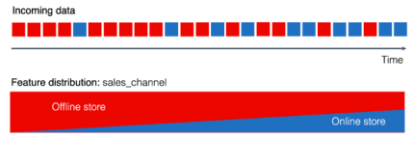


https://vitalflux.com/difference-between-online-batch-learning/

  - The models must be trained using all the available data every single time.
    - This approach may take a lot of time and computing resources.
    - Typically done offline.
    - After the models are trained, they are launched (deployment phase)
      - They run without learning anymore.

Batch Learning

- Performance
  - Improving the model performance would require re-training all over again with the entire training data set.
  - Static models in nature
    - Once they get trained, their performance will not improve until a new model gets re-trained.
  - The model's performance tends to decay slowly over time
    - Data changes but the model remains unchanged: **data drift**
      - Solution: regularly retrain the model on up-to-date data and replace the old model.



https://www.evidentlyai.com/ml-in-production/data-drift

An example of data drift is a change in sales distribution by channel.

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Batch Learning

- Apply batch learning when:
  - The business requirements do not require frequent learning of models.
  - The data distribution is not expected to change frequently.
  - The resources (expertise) required for creating the system for incremental learning is not available.

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Online Learning

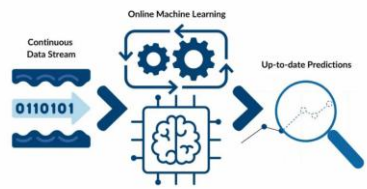
- The system is trained incrementally
  - Each learning step is fast
  - particularly well adapted for systems that receive data as continuous flow (e.g. stocks price) and need to adapt to changes in the data.
  - reduce the requirements of CPU/GPU, memory, ….



  - Main challenge
    - to assure permanently the quality of data, if bad data (e.g. malfunctioning sensor, …) is fed to the system its performance will gradually decline

Online Learning

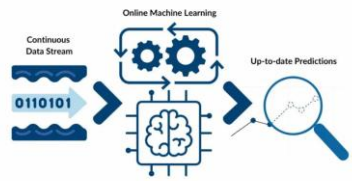
- Training is carried out in an incremental manner:
  - Continuously feeding data as it arrives in mini batches.
  - Each learning step is fast and cheap
  - The system can learn about new data on the fly
  - The old data doesn't need to be stored (save space / resources).
  - Can also be adapted to train models on large datasets.
    - The algorithm loads part of the data, runs a training step on that data and repeats the process through all data.



https://spotintelligence.com/2024/04/10/online-machine-learning/

Online Learning

- Learning rate
  - The rate at which the machine learning algorithm adapts to new data
    - critical issue
    - High learning rate → model will tend to forget the learning very quickly.
    - Low learning rate → model will be more like batch learning.



- Online models require only a single deployment in the production setting
  - They evolve over a period of time.

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Online Learning

- Disadvantages
  - If fed with bad data, the system will have bad performance
    - The impact is immediate.
  - It is very important to monitor closely the performance of the machine learning system.
  - Data governance needs to be implemented:
    - Data ingestion approach
    - ETL (extract, transform, load) pipelines
    - Feature extraction
    - …

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Online Learning

Main challenge: To assure permanently the quality of data!

Uber Eats Estimated Time of Delivery Model



To predict the total duration of this multi-stage process in real time, **Uber's data scientists have built regression models that require data from a variety of sources.**

These models depend on data from customer requests, historical information about restaurant performance, and real time information describing restaurant load, traffic conditions, and available drivers …

Uber **needs to monitor the distributions of the incoming data** to detect deviations that would cause models to act in unexpected ways.

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Machine Learning

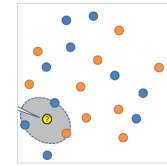
Types of Learning
Classification / Regression
Model based / Instance-Based Learning
Batch / Online Learning
Parameters / Hyperparameters

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Parameters vs Hyperparameters

- Parameters
  - Variables that define the model learned by the learning algorithm.
  - Directly modified by the learning algorithm based on the training data. $y = sign(\mathbf{w}^*\mathbf{x} + b^*)$
  - The goal of learning is precisely to determine the best values of these variables/parameters.

- Hyperparameter
  - It is a property of the learning algorithm, usually a numerical value.
  - That value has a high influence in how the algorithm works.
  - These parameters are not learned from data, they are set before running the algorithm
    - e.g. k nearest neighbor (the value of *k* – number of neighbours)
    - They are absolutely critical in complex models (ANN, CNN, etc.)

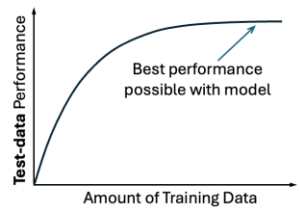


Machine Learning

Types of Learning
Classification vs Regression
Model based vs Instance-Based Learning
Batch vs Online Learning
Main Challenges

Main Challenges

- Insufficient Quantity of Training Data
  - Even for relatively simple problems thousands of instances may be required to properly address the training
    - Complex problems (e.g. image recognition) may require millions of examples.
    - A common rule is to have **at least** 10 times more training data than the number of features.
  - The typical evolution of the classifiers performance according to the dimension of training data set.

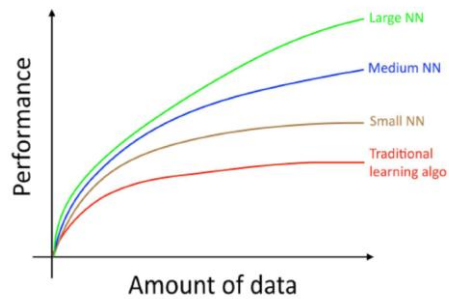


  - Problem:
    - Frequently **it is not easy or cheap to get extra training data**, so small and medium sized datasets are very common in real machine learning applications

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Main Challenges

- Insufficient Quantity of Training Data
  - Models applied to the classification process **have different abilities** to deal with the amount of data (i.e. the amount of data cannot be analyzed isolated)



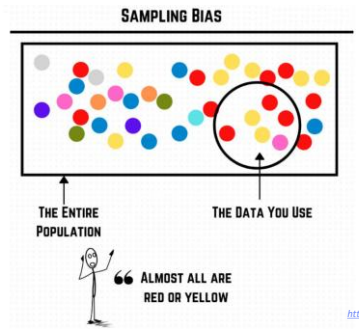
" … even as you accumulate more data, usually the performance of older learning algorithms, such as logistic regression, … its learning curve "flattens out," and the algorithm stops improving even as you give it more data…"

Machine Learning Yearning, Technical Strategy for AI Engineers in the Era of Deep Learning, Andrew NG, 2018

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Main Challenges

- Nonrepresentative Training Data
  - The training data should be representative of the new cases.
    - This is a key aspect to assure the generalization capability of the classifier, i.e. classifier has a good performance with new instances
    - If the sample **is too small it can originate sampling noise** (nonrepresentative data as a result of chance) or **even large samples can be nonrepresentative if the sampling method was biased** (sampling bias)



https://productiveclub.com/selection-bias/

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Main Challenges

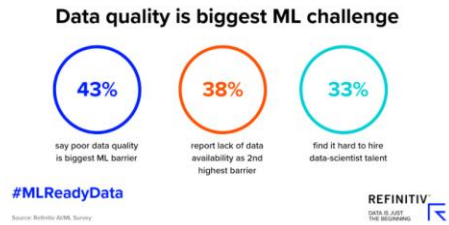
- Poor Quality Training Data
  - Frequent problems:
    - errors
    - outliers
    - noise (poor quality measurements)
    - missing data
    - irrelevant features
    - …
  - In real applications training data must be cleaned up
    - Critical Operation!!



http://solutions.ait.ac.th/garbage-in-garbage-out/

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Main Challenges



https://www.refinitiv.com/perspectives/ai-digitalization/machine-learning-for-risk-and-returns/

- Clean up (some examples):
  - If some instances are **clearly outliers**, probably they **should be discarded or properly handled**
  - If some instances have **more than 5% missing features**, probably they should be ignored or an imputation data should be carried out.



  - If a **feature is missing in several instances**, probably that feature should be evaluated.
    - Assessment of model performance with and without that specific feature …

Main Challenges

- Irrelevant Features



http://solutions.ait.ac.th/garbage-in-garbage-out/

  - The system will only be able to learn if the training data contains enough **relevant features** and **not too many irrelevant ones**.
  - Assure a good set of features is a critical step. It is designated as **feature engineering** and may comprise the following:
    - Feature Selection
      - Select the most useful features to train among the set of all features
    - Feature Extraction
      - Combine existing features to produce a more useful one
    - New Features
      - By gathering new data

Main Challenges

- Imbalanced Dataset
  - The number of observations per class is not equally distributed;
  - Often there is a large amount of data/observations for one class (**majority class**), and much fewer observations for one or more other classes (**minority classes**).
    - Imbalanced ratio ranges from 1:4 up to 1:100
    - Extremely imbalanced datasets (frequent in real-life applications, e.g. credit fraud) may vary from 1:1000 up to 1:10000

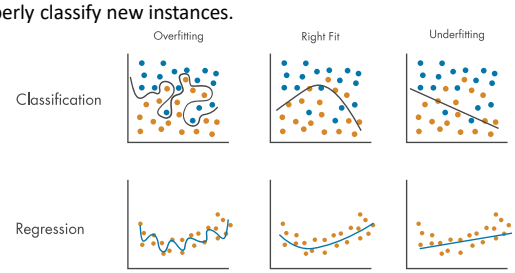


https://chrisherd.medium.com

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Main Challenges

- Overfitting Training Data
  - Model performs well on the training data but it **does not generalize well** (bad performance in new instances)
    - In spite of the good results with the training dataset the classifier is USELESS, as it cannot properly classify new instances.



https://www.mathworks.com/discovery/overfitting.html

Error	Overfitting	Right Fit	Underfitting
Training	Low	Low	High
Test	High	Low	High

  - Very common problem!!
    - It is very common in complex models (e.g. deep learning models). Some techniques to reduce the risk of overfitting should be applied.

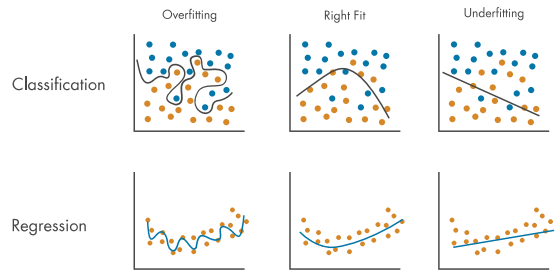
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Main Challenges

- To avoid overfitting some alternatives may be considered:
  - Simplify the model
    - reduce the number of parameters, number of attributes, constrain the model (e.g reduce the degrees of freedom of a linear model, …)
  - Gather more training data
  - Reduce noise in training data
    - e.g. clean up data, remove outliers, …
  - Define constrains based on hyper-parameters
    - Their value is initialized before the learning takes place
      - e.g. Decision Tree Algorithm: Total number of leaves in the tree; height of the tree, minimum leaf split size; number of leaves in each level of the tree.

Main Challenges

- Underfitting Training Data
  - Occurs when the model is too simple to learn an underlying structure of the data.
  - Main strategies to overcome this problem:
    - Select a more powerful model, with more parameters
    - Better set of features
    - Reduce the constraints of the model

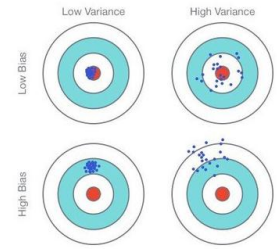


Main Challenges

- error

$$error = reducible\ error + irreducible\ error$$

  - Reducible error
    - bias
      - The difference between the average prediction of the model and the correct value.
      - A model with high bias always leads to high error on training and test data.
      - High bias suggests an : **underfitting situation**!
    - variance
      - is the amount that the estimate of the target function will change if different training data was used.
      - High variance means that the algorithm is strongly influenced by the specificities of the training data.
      - High variance suggests an **overfitting situation.**



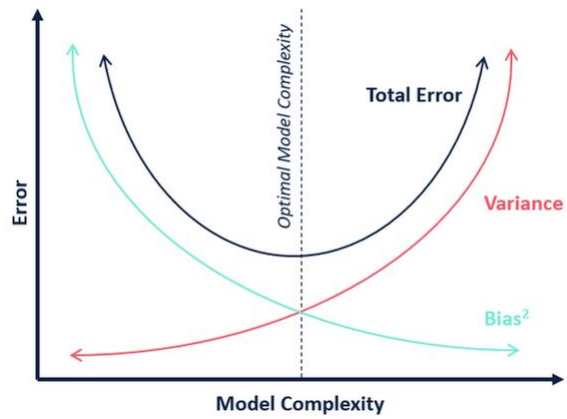
$$reducible\ error = bias^2 + variance$$

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Main Challenges

- Bias-Variance Tradeoff
  - Optimal Model Complexity

$$reducible\ error = bias^2 + variance$$

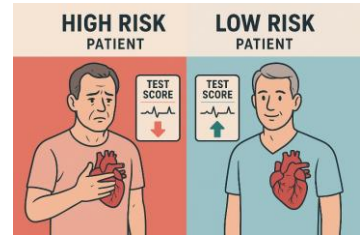


https://adolfoeliazat.com/2020/07/21/bias-variance-tradeoff-in-machine-learning/

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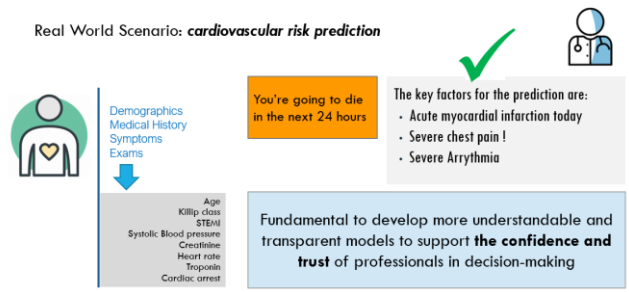
Main Challenges

- Trust
  - In areas with significant user impact the effective application of ML models depends on **Trust!**
    - Example: support to the clinical decision where the cost of ML model misclassification is potentially high
      - Predict the risk of:
        - a sepsis
        - an hospital readmission
        - a cardiac event
        - …



Main Challenges

- Trust
  - Related with several concepts:
    - Interpretability
    - Explainability
    - Reliability
    - Fairness



Jorge Henriques, *Explainable and Reliable machine learning Models*, TRUST WS, ICDM 2024, Abu Dhabi, 2024.

  - This is a very active area of investigation among AI researchers

Main Challenges

- Trust
  - Essential from different perspectives:

Data Scientist	End User	Policy makers
▪ What are the most important inputs/features? ▪ Which model should be selected ? ▪ Why does the model not provide correct answers ? ▪ How to improve the model?	▪ Can I trust on these results? ▪ What and how can I change the behavior for a more positive outcome? ▪ How to communicate the results ?	▪ Does the model make fair decisions ? ▪ Who is the responsible ?

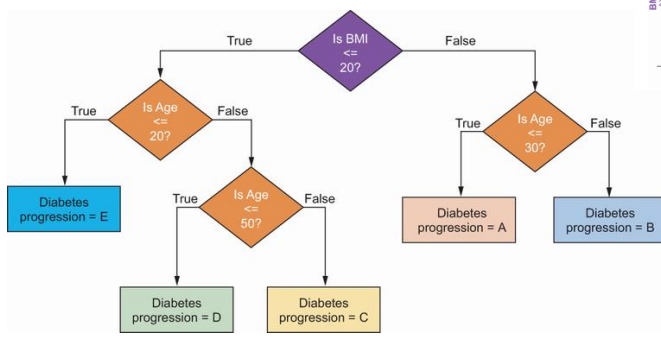
Main Challenges

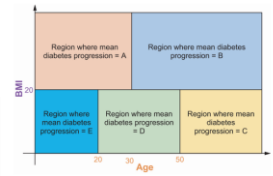
- **Interpretability**
  - It can be defined as the ability to describe how a model works in understandable terms to a human
    - i.e. provide descriptions for ML models' predictions in terms that humans can easily understand.
      - How does the model work?
        - Simple models / white box models
        - Lack of performance

IF temperature > 38°C and white blood cell count > 12000 THEN pneumonia.

Main Challenges

- Interpretability intends to describe the effect of the inputs in the output of the model
  - White-box models



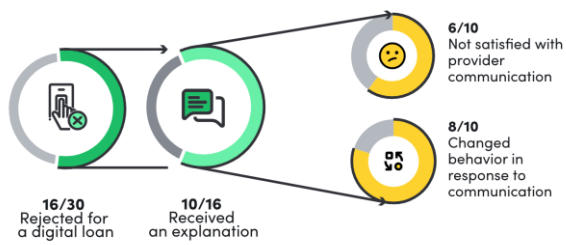


https://livebook.manning.com/book/interpretable-ai/chapter-2/70

  - Frequently the answer to this question is not possible as there are a high number of features involved and their calculations can be complex

Main Challenges

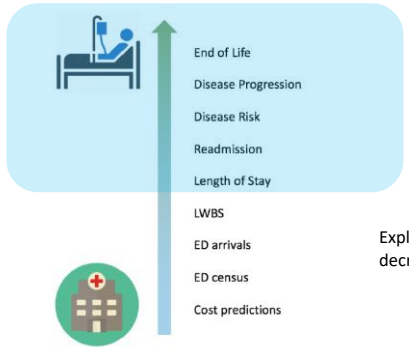
- Explainability
  - Right to Explanation
    - There is also increasing societal pressures and legislation requiring **that AI predictions are explained to users,** such as the European Union General Data Protection Regulation, who recently approved the **"right to explanation"** for any organization which use ML algorithms that make decisions with **significant user impact** (loan, health care, health insurance, recruitment, …)



https://www.centerforfinancialinclusion.org/trust-of-data-usage-sources-and-decisioning-perspectives-from-rwandan-mobile-money-users

Main Challenges

- Right to Explanation



Healthcare Use Case

The arrow represents the increasing need of explanations.

Context is critical when considering interpretability.

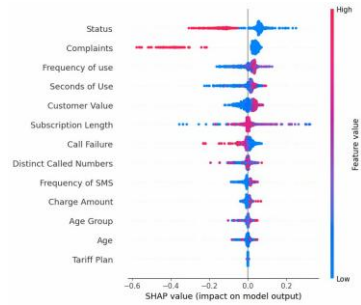
From the **Length of Stay** to **End of Life:**
Explanations will be helpful in clinical decision making, while a slight decrease in model performance might be tolerated

Trade-off between Performance vs Transparency

https://www.turing.ac.uk/research/impact-stories/a-right-to-explanation

Main Challenges

- Explainability
  - An explanation can be evaluated in two ways: according to its **interpretability**, and according to its **completeness**
    - Explainability answers to **Why**
      - Why did the model make this specific prediction?
    - Several Approaches: Surrogate models, Shappley values, visualization.

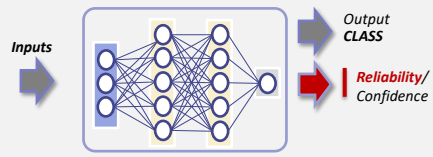


https://www.datacamp.com/tutorial/introduction-to-shap-values-machine-learning-interpretability

Main Challenges

- Reliability
  - Pointwise Reliability

- Is it possible to compute an **individual reliability measure** for a specific input (instance) ?
  - **Additionally** to the output, provide a pointwise confidence estimation for the output.
  - Fundamental for the **trustiness** of the user

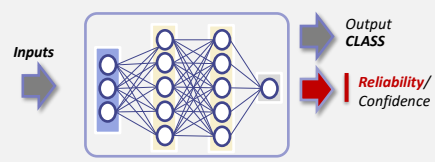


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Main Challenges

- Reliability
  - What means an accuracy of 85% ?
  - It is known that 85% (in average) of the decisions are correct
  - It is a global metric.
  - What does it mean for a particular decision?
    - ?
  - How can we rely/trust in a particular decision of a ML model?
    - ?

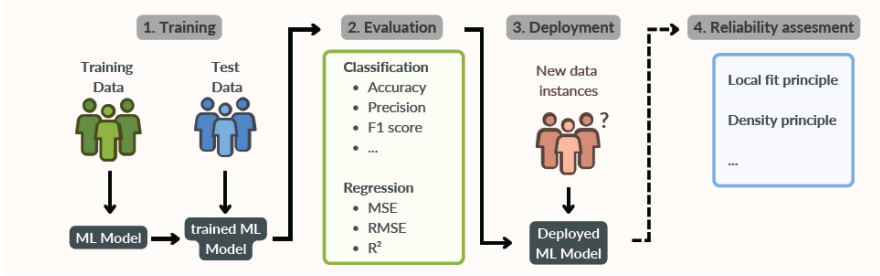
- Is it possible to compute an **individual reliability measure** for a specific input (instance) ?
  - **Additionally** to the output, provide a pointwise confidence estimation for the output.
  - Fundamental for the **trustiness** of the user



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Main Challenges

- Reliability
  - the probability that the predict class is equal to the true class

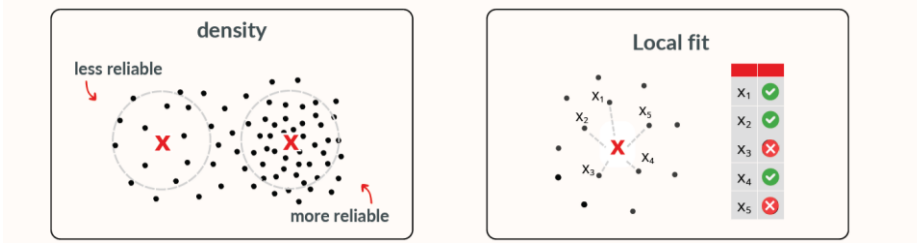


Jorge Henriques, *Explainable and Reliable machine learning Models*, TRUST WS, ICDM 2024, Abu Dhabi, 2024.

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Main Challenges

- Reliability
  - **Density**
    - Uses distance measures to compare the instance under interest to the examples present in the training set
  - **Local fit**:
    - assesses how the trained model performs on training subsets nearest to the instance under evaluation to derive the confidence measures.
    - Basically, the AI model will perform well (is **reliable**):
      - On instances analogous to the training instances that exist on the data set
      - Where the model performs correctly.



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Machine Learning

Types of Learning
Classification vs Regression
Model based vs Instance-Based Learning
Batch vs Online Learning
Main Challenges
Validation

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Main Challenges

- Generalization:
  - the model's ability to adapt properly to new, previously unseen data, drawn from the same distribution as the one used to create the model.

    How well the model will perform on instances it has never seen before.

- Validation
  - The only way to assess how well a model will generalize to new cases is try it out on new cases.

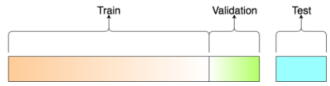


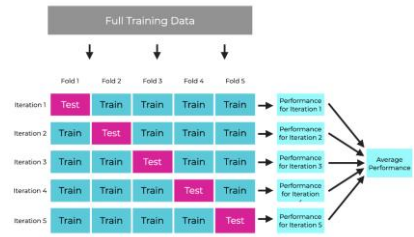
  - **Important:** The test set must be as representative as possible of the data.

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Main Challenges

- Validation
  - Split the dataset into two different sets:
    - **training dataset** (train the model)
    - **test set** (test the model)
      - Usually 70%/30% or 80%/20%, however there are other approaches to **reinforce validation** (*e.g.* Cross Validation, …)



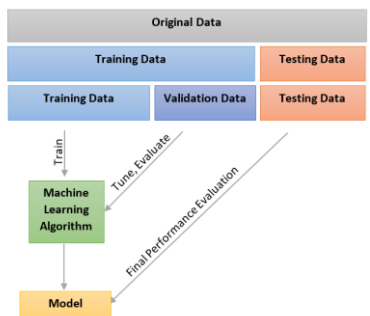


      - The error rate on new cases (generalization error) is obtained based on test dataset.
        - If the training error is low but the generalization error is high (overfitting situation: avoid!)

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Main Challenges

- Train-test split Validation
  - A validation set can also be applied, as an intermediate validation (tuning model changing the hyper parameters) before submitting the model to the test set.
    - The validation dataset is important to avoid a biased evaluation of the model's performance



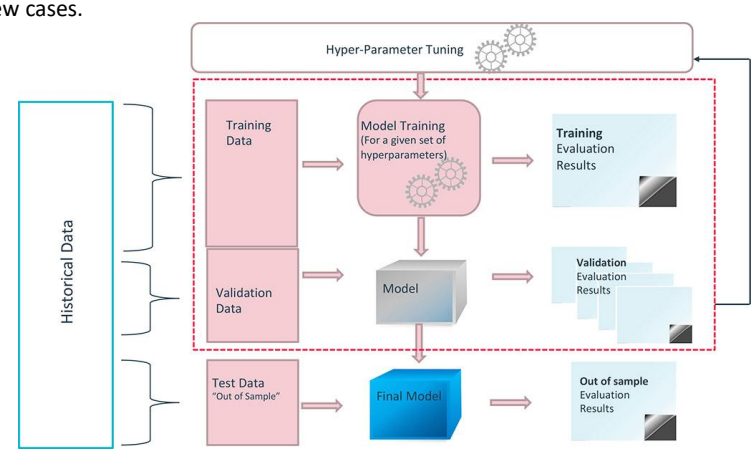
https://medium.com/free-code-camp/how-to-get-a-grip-on-cross-validations-bb0ba779e21c

- Training dataset is used to train a few candidate models
- Validation dataset is used to evaluate the candidate models (tunning of the model)
- One of the candidates is chosen
- The chosen model is trained with the entire training dataset
- The trained model is evaluated with the test dataset (generalization)

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Main Challenges

- Validation
  - The only way to assess how well a model will generalize to new cases is try it out on new cases.



https://www.ram-ai.com/machine-learning/

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Mathematical Notation

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Mathematical Notation

- scalar
  - simple numerical value x, a
- vector
  - ordered list of scalar values (attributes) $\mathbf{x}, \mathbf{w}$
  - An attribute of a vector is w_i, x_i or $w^{(j)}, x^{(j)}$

where i, j denotes a specific dimension/position of the vector.

$$\mathbf{x} = [2 \;\; 3] \qquad x_1, x_2 \,?$$

$$x_1 = 2, x_2 = 3$$

- A variable can have two or more indices $x_i^{(j)}, x_{ij}^{(u)}$
- j shouldn't be confused with the power operator, in this case the notation is $(x^{(j)})^2$

Mathematical Notation

- Matrix
  - An $r \times c$ matrix is a rectangular array of elements with r rows and c columns.
    - represented by capital letters (bold) $\mathbf{A}, \mathbf{W}$

$$\mathbf{A} = \begin{bmatrix} 2 & 5 & -3 \\ 21 & -5 & -1 \end{bmatrix}$$

- Set
  - unordered collection of unique elements
    - denoted by handwriting capital letter $\mathcal{S}, \mathcal{T}$
    - can be finite $\mathcal{S} = \{1, 2, 5, 6, 8\}$
    - can be infinite and includes all values in some interval $\mathcal{S} = [2, 5]$
    - can include all values in some interval excluding the limits $\mathcal{S} = (1,4)$; $\mathcal{S} =]1,4[$
    - $\mathbb{R}$ includes all numbers from minus to plus infinity

Mathematical Notation

- set
  - element x belongs to a set $\mathcal{S}$ is denoted by $x \in \mathcal{S}$
  - a set $\mathcal{S}_3$ obtained from the intersection of $\mathcal{S}_1$ and $\mathcal{S}_2$ is represented as $\mathcal{S}_1 \cap \mathcal{S}_2 \rightarrow \mathcal{S}_3$
  - a set $\mathcal{S}_3$ obtained from the union of $\mathcal{S}_1$ and $\mathcal{S}_2$ is represented as $\mathcal{S}_1 \cup \mathcal{S}_2 \rightarrow \mathcal{S}_3$
    - e.g.

$$\{1,3,5,8\} \cap \{1,8,4\} \rightarrow \ ? \quad \{1,8\}$$
$$\{1,3,5,8\} \cup \{1,8,4\} \rightarrow \ ? \quad \{1,3,4,5,8\}$$

- Capital Sigma Notation

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + x_3 + \ldots + x_n$$

Mathematical Notation

- Capital Pi Notation
  - a product of elements in a collection of attributes of a vector

$$\prod_{i=1}^{n} x_i = x_1 \cdot x_2 \cdot x_3 \cdot \ldots \cdot x_n$$

- Operations on sets
  - A derived set creation operator like:

$$\mathcal{S}' \leftarrow \{x^2 \mid x \in \mathcal{S}, x > 3\}$$

where a new set $\mathcal{S}'$ is created through x^2 such that x belongs to $\mathcal{S}$ and x is greater than 3

Mathematical Notation

- Operations on Vectors
  - considering: $\mathbf{x} = [x_1 \;\; x_2 \ldots x_m]; \mathbf{z} = [z_1 \;\; z_2 \ldots z_m]$
  - Sum
    - $\mathbf{x} + \mathbf{z} = [x_1 + z_1 \quad x_2 + z_2 \quad \ldots \quad x_m + z_m]$
  - Difference
    - $\mathbf{x} - \mathbf{z} = [x_1 - z_1 \quad x_2 - z_2 \ldots \quad x_m - z_m]$
  - A vector multiplied by a scalar
    - $c\mathbf{x} = [cx_1 \quad cx_2 \ldots cx_m]$
  - A dot product of two vectors is a scalar (vectors must have the same dimensionality)
    - $\mathbf{wx} = \sum_{i=1}^{m} w_i x_i$

Mathematical Notation

- Operations with vectors and matrices
  - if the vector is multiplied on the right of the matrix, the vector has to have the same number of rows as the number of columns of the matrix.

$$\mathbf{Wx} = \begin{bmatrix} w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} w_{1,1}x_1 + w_{1,2}x_2 + w_{1,3}x_3 \\ w_{2,1}x_1 + w_{2,2}x_2 + w_{2,3}x_3 \end{bmatrix} = \begin{bmatrix} \mathbf{w}^{(1)}\mathbf{x} \\ \mathbf{w}^{(2)}\mathbf{x} \end{bmatrix}$$

- Transpose of a vector / matrix
  - Transforms rows/columns into columns/rows
    - $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ then $\mathbf{x}^T = [x_1 \quad x_2]$

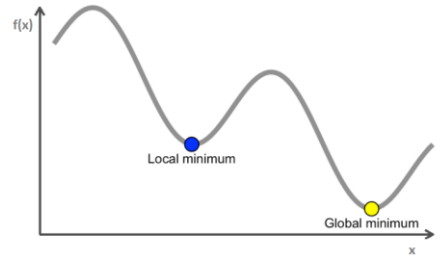
$$\mathbf{x}^T\mathbf{W} = [x_1 \quad x_2] \begin{bmatrix} w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \end{bmatrix} = [x_1 w_{1,1} + x_2 w_{2,1} \quad x_1 w_{1,2} + x_2 w_{2,2} \quad x_1 w_{1,3} + x_2 w_{2,3}]$$

Mathematical Notation

- Functions
  - associates each element x of a set $\mathcal{X}$ (domain) to a single element y that belongs to $\mathcal{Y}$ the codomain of the function.

$$y = f(x)$$

  - $f(x)$ has a local minimum at $x = c$ if $f(x) \geq f(c)$ for some open interval around $x = c$
  - The minimal value among the local minima is called the global minimum.



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Mathematical Notation

- max ; arg Max
  - Given a set of values $\mathcal{A} = \{a_1, a_2, \ldots, a_n\}$:
    - The operator $max_{a \epsilon A} f(a)$ returns the highest value $f(a)$ for all elements in the set $\mathcal{A}$
    - The operator $arg\ max_{a \epsilon A} f(a)$ returns the element of the set $\mathcal{A}$ that maximizes $f(a)$
    - when the set is implicit or infinite the notation can be simplified:
      - $max_a f(a)$; $arg\ max_a f(a)$
    - Operators min ; $arg\ min$ work in a similar way.

- Assignment Operator
  - The expression $a \leftarrow f(x)$ means that a gets assigned a new value of $f(x)$
    - e.g. $\mathbf{a} \leftarrow [a_1, a_2]$ the variable **a** gets the two-dimensional vector $[a_1, a_2]$

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Mathematical Notation

- Derivative
  - f' of f describes how fast f increases or decreases.
    - If f' is a constant value then the function increases (decreases) constantly at any x of its domain.
    - If f' is a function then a function can increase at a different pace in different regions of its domain
    - if $f' = 0$ the function is constant
  - Some basic derivatives:
    - $f(x) = x^2$
    - $f(x) = 6x$?
    - $f(x) = 2$
  - Chain rule (composite function when a function is applied to the result of other function)
    - $F(x) = f\big(g(x)\big) \rightarrow F'^{(x)} = f'(g(x))g'(x)$

$$F(x) = (5x+1)^2 \rightarrow \quad g(x) = (5x+1) \; ; f\big(g(x)\big) = (5x+1)^2$$

$$F'(x) = 2(5x+1)g'^{(x)} = 2(5x+1)5 = 50x + 10$$

Mathematical Notation

- Gradient
  - It is a generalization of derivative for functions considering several input variables
    - Thus, it is a vector of partial derivatives.
    - A partial derivative can be found considering all the other inputs as constant values

$$f(x_1, x_2) = ax_1 + bx_2 + c$$

The partial derivative of f with respect to x_1 is given by:

$$\frac{\partial f}{\partial x_1} = a + 0 + 0 = a$$

with respect to x_2 is given by:

$$\frac{\partial f}{\partial x_2} = 0 + b + 0 = b$$

The gradient of f, is denoted and given by:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix}$$